

CISCO VIRTUAL INTERNSHIP 2026

Secure Hybrid Data Center Network Architecture & Multi-Cloud Workload Protection

Program & Authority:	Cisco Virtual Internship 2026 / AICTE & Cisco Networking Academy
Problem Statement:	Cyber Security: Enterprise Hybrid Data Center & Workload Security
Key Technologies:	Cisco ASA/Firepower, Cisco ISE, Cisco Duo, AWS TGW, Calico CNI, Istio mTLS
Submission Deadline:	30th August 2026

1. Executive Summary

Modern enterprise and higher education institutions are transitioning from isolated on-premises data centers to distributed hybrid multi-cloud architectures. Today's academic and enterprise workloads span private data centers (Red Hat OpenShift, local database clusters) and public cloud environments (AWS EKS, RDS, Azure AKS). While this hybrid paradigm unlocks unprecedented scalability and collaborative power, it fundamentally dissolves the traditional network perimeter.

This project presents an enterprise-grade Secure Hybrid Data Center Network Architecture designed to balance simplicity, rock-solid security, and linear scalability. Leveraging Cisco enterprise security solutions (Cisco ASA 5506-X / Firepower, Cisco ISE, Cisco Duo MFA, Cisco Stealthwatch) combined with cloud-native controls (AWS Transit Gateway, SG-to-SG referencing, Calico CNI default-deny, and Istio mTLS), this architecture guarantees that any breach within a single container, virtual machine, or remote endpoint is immediately contained, logged, and isolated.

2. Core Technical Objectives

- **Hybrid Interconnect Protection:** Secure site-to-site IPsec VPN over BGP with IKEv2 / AES-256-GCM and Cisco ASA stateful deep inspection.
- **Unified Identity & Access Management (IAM):** Eliminate static credentials via OIDC federation (EKS IRSA) and Least-Privilege RBAC.
- **Multi-VPC Micro-Segmentation:** Hub-and-Spoke Transit Gateway topology with 3-tier subnets and SG-to-SG referencing.
- **Kubernetes & Microservice Hardening:** Calico CNI default-deny network policies, Restricted Pod Security Standards, and Istio mTLS 1.3.
- **Secure Remote Faculty Access (ZTNA):** Cisco AnyConnect SSL VPN with Cisco Duo MFA and dynamic Cisco ISE posture validation.
- **DevSecOps Governance:** Automated CI/CD security gates (SAST, Checkov IaC, Trivy container scanning, Cosign signing).

3. Network Topology & IP Addressing Allocation

Segment Name	Subnet CIDR	VLAN / Zone	Security Function
Enterprise Core Transit	10.10.0.0/30	P2P	Core Router to ASA Inside link
Faculty Workstations	10.10.10.0/24	VLAN 10	On-Premises Faculty Desktops (802.1X)
Student & Staff LAN	10.10.20.0/24	VLAN 20	General Campus; DB access blocked

On-Premises OpenShift/DC	10.10.30.0/24	VLAN 30	Private DC nodes & DB servers
Network Management	10.10.99.0/24	VLAN 99	Admin jumpboxes & out-of-band SSH
Faculty AnyConnect Pool	172.16.50.0/24	SSL VPN	Dynamic IP pool for remote ZTNA faculty
Enterprise WAN Edge	203.0.113.0/30	Outside (0)	Cisco ASA Public IP (203.0.113.2)
Cloud Transit WAN Edge	198.51.100.0/30	Cloud WAN	Cloud Transit Gateway (198.51.100.2)
Cloud Tier 1 (ALB Ingress)	172.20.1.0/24	Public	AWS Application Load Balancers & WAF
Cloud Tier 2 (EKS App)	172.20.2.0/24	Private	Kubernetes workers; zero public IPs
Cloud Tier 3 (Isolated DB)	172.20.3.0/24	Isolated	Managed DB (RDS); zero internet routing

4. Cloud Security Groups & IAM Architecture

Stateful Security Groups (SG-to-SG Referencing): Rather than relying on fragile CIDR blocks, cloud security groups reference peer Security Group IDs. Ingress to the EKS microservices tier is permitted strictly from sg-prod-ingress-alb on port 8080. Database ingress is strictly permitted ONLY from sg-prod-eks-microservices on port 5432, preventing rogue instances or compromised containers in other subnets from probing the persistence layer.

IAM Roles for Service Accounts (IRSA): Pods running in AWS EKS assume temporary, scoped IAM roles via OIDC Web Identity Federation. Applications never store static access keys, eliminating secret leakage risks.

5. Kubernetes & Remote Faculty Access Security

- **Calico CNI Default-Deny:** Global default-deny network policies enforce zero-trust pod communication.
- **Istio mTLS 1.3:** Mutual TLS with cryptographic SPIFFE identities for all pod-to-pod streams.
- **Pod Security Standards:** Restricted profile (non-root execution, read-only root filesystems, drop all capabilities).
- **Cisco AnyConnect ZTNA:** Remote faculty access protected by Cisco Duo MFA push notifications and Cisco ISE dynamic endpoint posture checks (OS patch, disk encryption, EDR health).

6. Architectural Balance: Simplicity, Security & Scale

The architecture successfully balances operational simplicity (Transit Gateway hub, GitOps automation, unified SSO), uncompromising security (defense-in-depth, zero-trust micro-segmentation, continuous telemetry with Cisco Stealthwatch), and cloud-scale elasticity (supporting thousands of microservices and up to 50 Gbps per VPC attachment).